

Orthogonal Projection - Revised.

Let V be an inn. prod space, (not necessary finite dimension!) and $W \subseteq V$ be a subspace.

For any vector $\beta \in V$, if $\alpha \in W$ satisfies that:

$$\|\beta - \alpha\| \leq \|\beta - \gamma\| \quad \text{for all } \gamma \in W \quad \dots (K)$$

Then, the α is called best approximation to β by W , simply b.app.

Thm. $\alpha \in W$ is b.app to $\beta \in V$ by W iff for all $\mu \in W$, $\langle \beta - \alpha, \mu \rangle = 0$.

Proof. For fixed $\alpha \in W$ and $\beta \in V$, for any $\gamma \in W$,

$$\|\beta - \gamma\|^2 = \|\beta - \alpha + \alpha - \gamma\|^2 = \|\beta - \alpha\|^2 + \|\alpha - \gamma\|^2 + 2 \cdot \operatorname{Re} \langle \beta - \alpha, \alpha - \gamma \rangle$$

Therefore, the condition (K) holds if and only if

$$0 \leq \|\alpha - \gamma\|^2 + 2 \cdot \operatorname{Re} \langle \beta - \alpha, \alpha - \gamma \rangle \quad \text{for all } \gamma \in W.$$

Now, take $\gamma = \alpha + \frac{\langle \beta - \alpha, \alpha - \mu \rangle}{\|\alpha - \mu\|^2} (\alpha - \mu)$ for any $\mu \in W$ s.t.

$$\text{Then, } 0 \leq \frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2} - 2 \cdot \frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2} = - \frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2}$$

And the inequality holds iff $\langle \beta - \alpha, \alpha - \mu \rangle = 0$, and all non zero vectors in W can be represented by $\alpha - \mu$, thus proved.

Conversely, if $\beta - \alpha$ is orthogonal to every vector in W , then $\operatorname{Re} \langle \beta - \alpha, \alpha - \gamma \rangle = 0$, thus $\|\beta - \gamma\|^2 - \|\beta - \alpha\|^2 = \|\alpha - \gamma\|^2 \geq 0$. $\square$

Thm. If b.app to $\beta \in V$ by W exists, then it is unique.

Proof. Suppose that $\alpha_1, \alpha_2 \in W$ satisfies for all $\gamma \in W$,

$$\langle \beta - \alpha_1, \gamma \rangle = \langle \beta - \alpha_2, \gamma \rangle = 0.$$

Then, take $\gamma = \alpha_1 - \alpha_2$, so $\langle \alpha_1 - \alpha_2, \alpha_1 - \alpha_2 \rangle = 0$. $\square$

Def. For all $\beta \in V$, there is b.app $\alpha \in W$, then the map

$$P: V \rightarrow V: \beta \mapsto \alpha$$

is called an Orthogonal Projection on W .

Thm. Let W be a fin. dim subspace of inn. prod space V .

Then, the Unique orthogonal projection on W exists. Further,

P is idempotent linear operator with $\text{Im } P = W$, $\text{Ker } P = W^\perp$.

Proof. Let $\{\alpha_1, \dots, \alpha_n\}$ be an orthonormal basis for W . Then, a map

$$P: V \rightarrow V: \beta \mapsto \sum_{i=1}^n \langle \beta, \alpha_i \rangle \cdot \alpha_i$$

1). It maps β to orthogonal projection by W .

For each $1 \leq i \leq n$,

$$\left\langle \beta - \sum_{i=1}^n \langle \beta, \alpha_i \rangle \alpha_i, \alpha_j \right\rangle = \langle \beta, \alpha_j \rangle - \langle \beta, \alpha_j \rangle = 0.$$

So, $\beta - P(\beta) \in W^\perp$ for all $\beta \in V$. (And, such a vector is unique, thus well-defined)

2). It is linear.

Let $\beta, \gamma \in V$ and $c \in F$ be given. Then,

$$P(c\beta + \gamma) = \sum \langle c\beta + \gamma, \alpha_i \rangle \cdot \alpha_i = c \cdot \sum \langle \beta, \alpha_i \rangle \cdot \alpha_i + \sum \langle \gamma, \alpha_i \rangle \cdot \alpha_i = cP(\beta) + P(\gamma).$$

3). It is idempotent.

Given $\beta \in V$, $P(\beta) \in W$. So, $P^2(\beta) = P(\beta)$. $\downarrow$ projection

4). $\text{Im } P = W$, $\text{Ker } P = W^\perp$.

$\text{Im } P \subseteq W$ is clear, and for any $\gamma \in W$, $P(\gamma) = \gamma \in \text{Im } P$. And,

$$\beta \in \text{Ker } P \Leftrightarrow P(\beta) = 0 \Leftrightarrow \langle \beta, \alpha_i \rangle = 0 \text{ for all } 1 \leq i \leq n \Leftrightarrow \beta \in W^\perp.$$

5). It is Hermitian.

Since $\text{Im } P \oplus \text{Ker } P = V$, let $x_1, y_1 \in \text{Im } P$ and $x_2, y_2 \in \text{Ker } P$ be given. Then

$$\langle P(x_1 + x_2), y_1 + y_2 \rangle = \langle x_1, y_1 + y_2 \rangle = \langle x_1, y_1 \rangle = \langle x_1 + x_2, y_1 \rangle = \langle x_1 + x_2, P(y_1 + y_2) \rangle.$$

□

Corollary. Let W be a fin. dim subspace of inn. prod space V ,

and let P be an orthogonal projection on W . Then, $I - P$ is an orthogonal projection on $W^\perp$.

Proof. Given $\beta \in V$, $\langle \beta - P(\beta), \alpha \rangle = 0$ for all $\alpha \in W$, or $\beta - P(\beta) \in W^\perp$. So $\text{Im}(I - P) = W^\perp$.

And, given $\gamma \in W^\perp$, $\beta - \gamma = P(\beta) + (\beta - P(\beta) - \gamma)$, therefore

$$\|\beta - \gamma\|^2 = \|P(\beta)\|^2 + \|\beta - P(\beta) - \gamma\|^2 \geq \|P(\beta)\|^2 = \|\beta - (\beta - P(\beta))\|^2 \text{ for all } W^\perp,$$

so $\beta - P(\beta)$ is the best approximation to β by vectors in $W^\perp$.

Note: let $E: V \rightarrow V$ be a projection. In general, E can not be determined by it's image.
 For example, let $V = \mathbb{R}^3$, $W_1 = \text{Span}\{(1, 0, 0)\}$, $W_2 = \text{Span}\{(0, 1, 0), (0, 0, 1)\}$, $W_3 = \text{Span}\{(0, 1, 0), (1, 1, 1)\}$.
 Then, $V = W_1 \oplus W_2 = W_1 \oplus W_3$, but $W_2 \neq W_3$. Moreover,
 $E: V \rightarrow V: (x_1, x_2, x_3) \mapsto (x_1, 0, 0)$, $E': V \rightarrow V: (x_1, x_2, x_3) \mapsto (x_1 - x_3, 0, 0)$
 Both E and E' are projections with $\text{Im } E = \text{Im } E' = \text{Span}\{(1, 0, 0)\}$, but $E \neq E'$,
 $\text{Ker } E = W_2$ and $\text{Ker } E' = W_3$.

Corollary. Let $\{\alpha_i, \dots, \alpha_n\}$ be an orthogonal set of non-zero vectors in inner prod space V . Then,
 for all $\beta \in V$, $\sum_{i=1}^n \frac{|\langle \beta, \alpha_i \rangle|^2}{\|\alpha_i\|^2} \leq \|\beta\|^2$, the equality holds iff $\beta = \sum \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i$.

Proof. Take $\gamma = \sum [\langle \beta, \alpha_i \rangle / \|\alpha_i\|^2] \cdot \alpha_i$. Then, $\langle \beta - \gamma, \gamma \rangle = 0$. So
 $\|\beta\|^2 = \|\gamma\|^2 + \|\beta - \gamma\|^2 \geq \|\gamma\|^2$.

Thm. Let W be a finite-dimensional subspace of an inn. prod space V .

If $P: V \rightarrow V$ is an orthogonal projection on W , that is,

$$\text{for any } \beta \in V, \quad \|\beta - P(\beta)\| \leq \|\beta - \gamma\| \text{ for all } \gamma \in W \quad \dots (*)$$

then, P is idempotent hermitian linear map s.t. $\text{Im } P = W$, $\ker P = (\text{Im } P)^\perp$, $V = \text{Im } P \oplus \ker P$

Proof. Since $P(\beta) \in W$, $P(P(\beta)) = P(\beta)$, for all $\beta \in V$, so $P^2 = P$, idempotent.

To show the linearity, let $\alpha, \beta \in V$ and $c \in F$ be given. Since $\alpha - P(\alpha), \beta - P(\beta) \in W^\perp$, so

$$c \cdot (\alpha - P(\alpha)) + (\beta - P(\beta)) = (c\alpha + \beta) - (c \cdot P(\alpha) + P(\beta)) \in W^\perp.$$

So, $cP(\alpha) + P(\beta) = P(c\alpha + \beta)$, linearity is obtained

To show the direct sum, given $\beta \in V$, $\beta = P(\beta) + (\beta - P(\beta))$, and since $\beta - P(\beta) \in W^\perp$, $V = \text{Im } P + W^\perp$ is obtained. And,

$$\alpha \in W^\perp \Leftrightarrow P(\alpha) = 0, \text{ because}$$

$$\alpha \in W^\perp \Leftrightarrow \forall \mu \in W, \langle \alpha, \mu \rangle = 0 \Leftrightarrow P(\alpha) = 0. \text{ So, } V = \text{Im } P + \ker P. \text{ And, } \text{Im } P \cap \ker P = \{0\},$$
$$\langle \alpha - 0, \mu \rangle = 0.$$

because $P(0) = 0$ and $\alpha = P(\alpha)$ implies $\alpha = 0$.

